Reinterpreting diffusive constraints: Concentration cloaking via homogenization and pseudoconformal mapping

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The manipulation of mass diffusion is crucial for applications spanning biotechnology to chemical engineering. However, established methods for designing diffusion metadevices are often constrained by high background diffusivity, inherent anisotropy, or implementation complexity. Here we present a unified theoretical framework for isotropic mass diffusion control that bridges transformation optics and convective transport. By applying homogenization theory to established convection-enhanced diffusion mechanisms, we demonstrate that a rotating fluid core effectively emulates the behavior of a near-zero-index medium in optics and thermotics. By combining this effective description with pseudoconformal mapping, we design an isotropic concentration cloak that remains applicable even in high-diffusivity environments. Numerical simulations validate that the cloak maintains robust performance for arbitrary target shapes in high-diffusivity regimes. Notably, the cloaked region remains sensitive to the external environment, allowing the internal concentration to track background changes in real time. A feasible experimental suggestion using perforated structures is also proposed. This work provides a diffusion-based reinterpretation of near-zero-index-inspired concepts and a synthesis of fluid mechanics and pseudoconformal mapping, offering a unified and practically accessible framework for engineering diffusive fields.

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I. INTRODUCTION

Mass diffusion [1–4] is a fundamental physical process in natural and engineered systems. It plays an important role in life sciences, environmental governance, and chemical engineering. Key applications such as targeted drug delivery, pollution control, and toxic gas protection pose two main scientific challenges in diffusion regulation [5–7]. First, one must minimize disturbance to the background concentration field. Second, precise and dynamic control of local concentrations is necessary. For example, liposomal drug delivery systems encapsulate therapeutic agents via concentric bilayer membranes [5]. This achieves a stealth effect that prevents premature clearance. Polymer coatings further promote immune evasion and maintain stable physiological concentration gradients, enhancing treatment efficacy. Similarly, pollutant leaks and the diffusion of toxic gases require the construction of safe zones [6]. These zones must maintain stable local concentrations and avoid affecting surrounding diffusion fields. Although concentration cloaks based on transformation theory theoretically achieve invisibility and responsive regulation in target regions [8], they depend heavily on anisotropic and inhomogeneous materials. While such properties can in principle be realized by layering different isotropic media, a more critical challenge arises: For all such passive designs, the required material parameters become physically unobtainable in prevalent high-diffusivity environments (see Appendix A). This fundamental limitation obstructs their path to practical implementation. To simplify fabrication, recent bilayer cloak designs use isotropic materials [9,10], offering a more convenient route for diffusion manipulation with homogeneous diffusivities.

However, in liquid media, natural materials typically exhibit a maximum diffusion coefficient of about $2 \times 10^{-9}\ \mathrm{m^2/s}$, which limits cloak design in high-diffusion backgrounds. Moreover, traditional bilayer cloaks lack an effective response within the cloaked region; specifically, the concentration inside this region evolves very slowly and is essentially decoupled from fluctuations in the background concentration. Active control methods, such as mobile robotic sources, improve environmental adaptability but significantly increase system complexity and energy consumption [11]. Thus, there is a pressing need for a diffusion-control strategy that circumvents anisotropic material constraints while delivering invisibility and responsiveness in high-diffusivity environments. Optical near-zero-index (NZI) materials have been widely applied in waveguide transmission [12–16], nonlinear optical enhancement [17], and radiation control [18]. In thermal systems, these materials are used to achieve invisibility and responsive control [19]. However, the mathematical analogy between NZI optics and convective mass transport remains underexplored. Unlike optical or thermal transport [20], diffusive mass transfer is intertwined with multicomponent coupling, chemical reactions, and fluid advection, which

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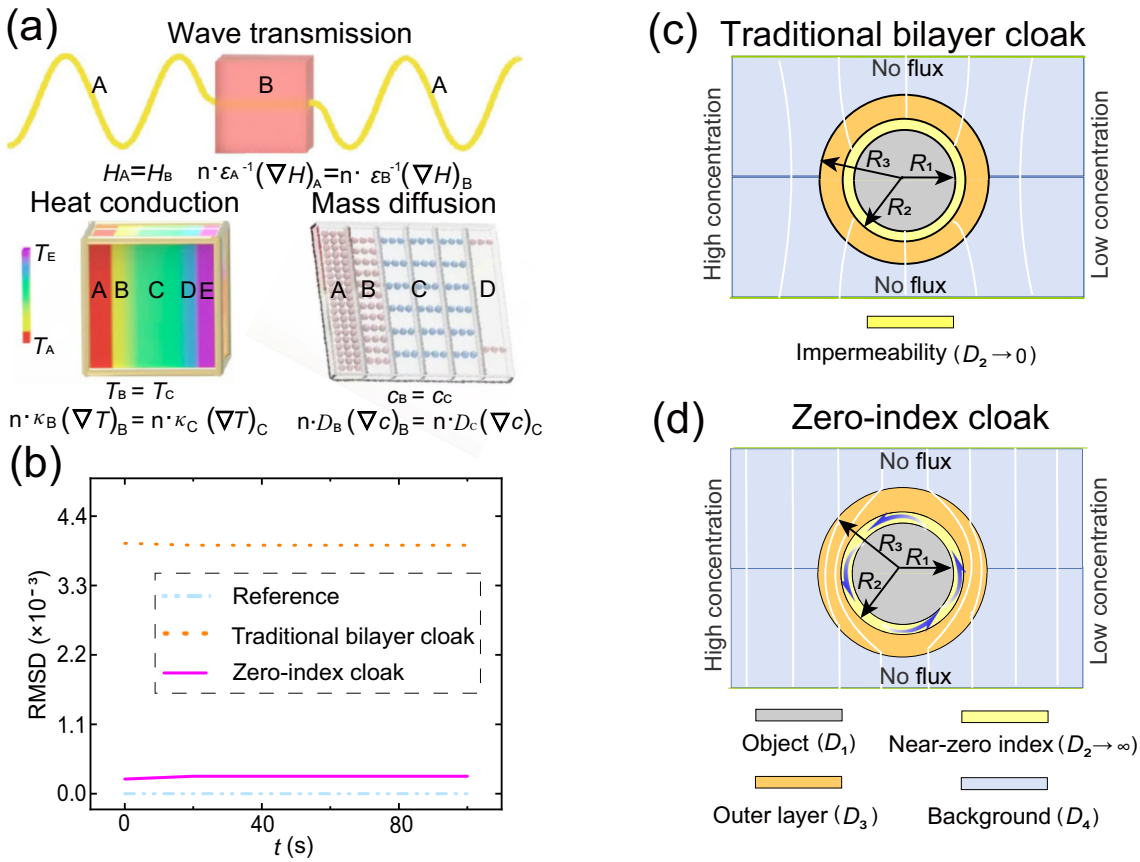

FIG. 1. Origin of a concentration near-zero-index (NZI) material and its effect in concentration cloaking. (a) Analogy between wave transmission, heat conduction, and mass diffusion. Shown on top is the electromagnetic wave transmission from a conventional material A to an optical NZI material B. The bottom left shows heat conduction through a thermal NZI material C connecting conventional materials A, B, D, and E. The bottom right shows mass diffusion through a concentration NZI material C connecting conventional materials A, B, and D. (b) Root-mean-square deviation (RMSD) of concentration over time for the reference structure, traditional bilayer cloak, and zero-index cloak, quantifying the degree of background concentration contour distortion. (c) Limitations of traditional bilayer cloaks in a high-diffusivity background $D_3 = (R_3^2 + R_2^2)D_4/(R_3^2 - R_2^2)$, which result in distorted concentration fields. (d) Zero-index cloak operating in a high-diffusivity background. The inner layer exhibits extremely high effective diffusivity achieved through rotational convection (blue arrows represent fluid motion), while the outer layer has a diffusivity lower than that of the background $D_3 = (R_3^2 - R_2^2)D_4/(R_3^2 + R_2^2)$, a condition achievable using natural materials.

complicates the realization of a concentration NZI medium. Closing this gap is compelling: A diffusion analog of the NZI medium would enable isotropic cloaks and background-responsive control in high-diffusivity liquids, motivating new theoretical frameworks for concentration NZI materials.

Here we establish a theoretical isomorphism between zero-index photonics, infinite thermal conductivity, and infinite mass diffusivity. We demonstrate that in the homogenized limit, a domain with effectively infinite diffusivity reproduces the wave-flattening effects of an optical NZI medium. Crucially, rather than seeking exotic new materials to achieve this extreme parameter, we revisit a classical fluid-mechanics mechanism: shear-driven rotational flow [21]. Through homogenization analysis, we prove that the convection-enhanced transport in such a flow mathematically maps to the required diffusive NZI constitutive parameters. This allows us to reinterpret the rotating fluid not merely as a mixer, but as a constitutive element in a transformation-based metamaterial device. This isotropy is essential for the cloaking function, as it allows the target region to equilibrate rapidly to a uniform concentration, enabling highly responsive local regulation. To preserve invisibility while retaining this responsiveness, we design a structure where an inner core mimicking NZI behavior is surrounded by an outer shell. This shell, designed using pseudoconformal mapping, cancels the perturbations that the active core would otherwise impose on the ambient concentration field, yielding an isotropic concentration zero-index-analog cloak. Numerical simulations across diverse target geometries in high-diffusivity backgrounds confirm both excellent invisibility and high sensitivity. Finally, we outline an experimental route that realizes effective concentration NZI behavior using perforated architectures interpreted via effective-medium theory, providing a practical platform for multiregion concentration control. The complete progression of our work, ranging from the fundamental concept to numerical validations and experimental proposals, is comprehensively illustrated throughout Figs. 1–7. This study provides a unified design strategy that fuses the geometric insights of transformation optics with the physical realizability of fluid convection [22,23], laying a solid theoretical and

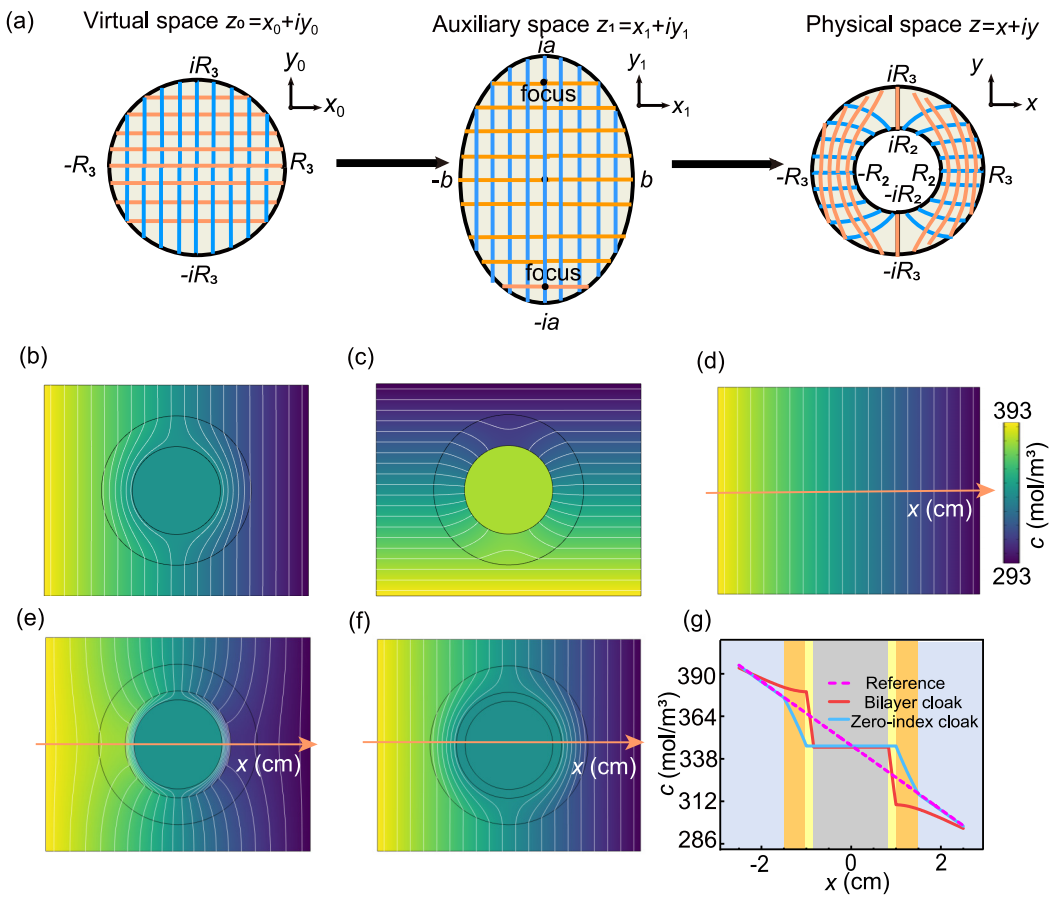

FIG. 2. Design of a concentration zero-index-analog cloak inspired by NZI concepts and pseudoconformal mapping theory. (a) Geometric transformation of the pseudoconformal mapping theory. (b) and (c) Physical duality of the pseudoconformal framework. For the same metadevice geometry and mapping, when a uniform concentration gradient is applied along the x axis, it functions as a zero-index cloak in (b), which corresponds to the primary configuration discussed in this work. When the gradient is applied vertically, the metadevice functions as a traditional bilayer cloak in (c). (d) Reference case without any object. (e) Simulation results for the traditional bilayer cloak operating in a high-diffusivity environment show that isoconcentration lines become distorted. (f) The zero-index cloak successfully achieves invisibility in the high-diffusivity background, maintaining undistorted isoconcentration lines. (g) Comparison of concentration values at various horizontal positions x for the reference, the traditional bilayer cloak, and the zero-index cloak, illustrating the improved cloaking performance of the zero-index cloak.

technical foundation for advanced applications such as biochemical regulation [24–26], pollution control, and industrial diffusion management [27].

II. THEORY

A. Diffusive analog to zero-index media

First, we consider the steady-state mass diffusion process without sources or sinks, in which the mass flux $\boldsymbol{M}$ must satisfy the continuity equation, which represents the conservation of mass,

$$\nabla \cdot \boldsymbol{M} = 0. \tag{1}$$

The mass flux is described by Fick's first law, which relates the flux to the concentration gradient,

$$\boldsymbol{M} = -D\nabla c, \tag{2}$$

where D is the diffusion coefficient (or diffusivity) and c is the concentration. By substituting Eq. (2) into Eq. (1), we obtain the governing equation for steady-state mass diffusion,

$$\nabla \cdot (D\nabla c) = 0. \tag{3}$$

This defines the Laplace-like governing equation for the concentration field. For the remainder of this work, we focus on a two-dimensional system in the x-y plane, assuming both the concentration field and material parameters are invariant in the z direction.

Cloaks designed using transformation theory or bilayer cloaks based on scattering-cancellation methods perform poorly in environments with high diffusivity. Specifically, designs based on transformation theory require extreme anisotropy, which is typically achieved by alternating layers of materials with high and low diffusivities. The bilayer cloak structure obtained via scattering cancellation is illustrated in Fig. 1(c), where the blue region denotes the background and the gray region denotes the object. The cloak occupies the annular domain $R_1 < r < R_3$, consisting of a yellow inner layer and an orange outer layer. Here R_1, R_2, and R_3 represent the radii of the object and the inner and outer layers

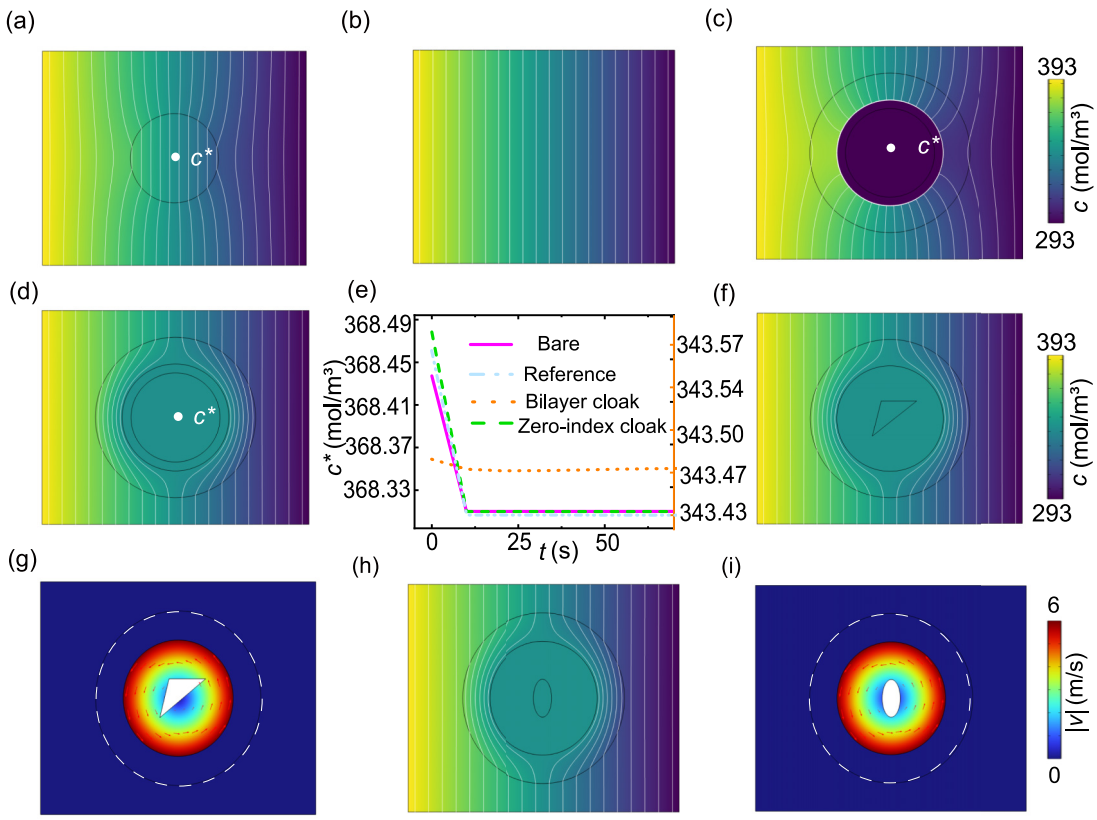

FIG. 3. Transient concentration distributions at $t = 500$ s and numerical simulation results for triangular and elliptical objects. (a) Bare object case, where white isoconcentration lines are distorted by the presence of the object. (b) Reference case, showing undistorted background concentration distribution. (c) The traditional bilayer cloak, whose outer layer has the same diffusivity as the background, still causes notable distortion of isoconcentration lines. (d) The zero-index cloak, featuring an outer layer with diffusivity lower than the background, effectively restores the background isoconcentration distribution. (e) Comparison of the time evolution of concentration at the geometric center of the cloaked object extracted from the cases in (a)–(d), quantifying the transient response. (f) and (h) concentration distributions and (g) and (i) velocity fields around the (f) and (g) triangular and (h) and (i) elliptical objects.

of the cloak, respectively, and D_1, D_2, D_3, and D_4 denote the diffusivities of these three regions and the background, respectively. The inner layer is impermeable, with $D_2 \to 0$, blocking concentration flux, while the outer layer preserves the background field by redirecting flux lines. Its diffusivity satisfies $D_3 = D_4(R_3^2 + R_2^2)/(R_3^2 - R_2^2)$. When $R_3 = 2R_2$, this simplifies to $D_3 = 5D_4/3$. For a water background ($D_4 \approx 2 \times 10^{-9}$ m^2/s), the required outer-layer diffusivity is 3.33×10^{-9} m^2/s, which exceeds the diffusivities of most natural solutes. This implies that strictly passive scattering-cancellation cloaks face a fundamental material bottleneck in aqueous environments.

Here we design a cloak that operates in a zero-index-analog regime to overcome this limitation, sharing the same bilayer structure as traditional cloaks [see Fig. 1(d)]. Inspired by optical and thermal NZI materials, we introduce an inner layer operating in a zero-index-analog regime [Fig. 1(a)]. While the NZI material tends to disturb the external concentration field, these disturbances are fully compensated by an outer shell with analytically determined moderate diffusivity via pseudoconformal mapping theory [28], as demonstrated in Fig. 1(d).

We briefly review the physical meaning of NZI materials in optics and thermal systems. In the frequency domain, the Helmholtz equation for electromagnetic fields $(\nabla^2 + \mu\varepsilon\omega^2)\mathbf{E} = 0$, where ω is angular frequency and ε and μ are the permittivity and permeability of the material, respectively, is similar in form to the Fourier-transformed heat equation $(\nabla^2 - i\omega\rho c_p/\kappa)T = 0$, where ρ is the density, c_p is the heat capacity, κ is the thermal conductivity, and T is the temperature. When the index $n = \sqrt{\mu\varepsilon} \to 0$ or the thermal diffusion parameter $\rho c_p/\kappa \to 0$, both reduce to Laplace's equation, yielding uniform phase or temperature distributions. Field continuity across NZI material interfaces requires that the curls of electromagnetic fields vanish as $\mu \to 0$ and $\varepsilon \to 0$, enforcing spatial uniformity. Analogously, in thermal NZI materials, the continuity of temperature and normal heat flux at the interface, combined with $\kappa \to \infty$, leads to a vanishing temperature gradient inside the NZI material region. These cases demonstrate that the NZI limit effectively enforces uniform internal fields. This establishes a rigorous isomorphism: The infinite-diffusivity limit in mass transport is the direct diffusive analog to the zero-index limit in wave physics.

The steady-state source-free diffusion process satisfies $\nabla \cdot (D\nabla c) = 0$, where D is the diffusivity and c is the concentration. Mathematically, under the limit $D \to \infty$, the diffusion equation degenerates to $\nabla^2 c = 0$ with the boundary condition $\nabla c \to 0$, enforcing a uniform concentration field throughout

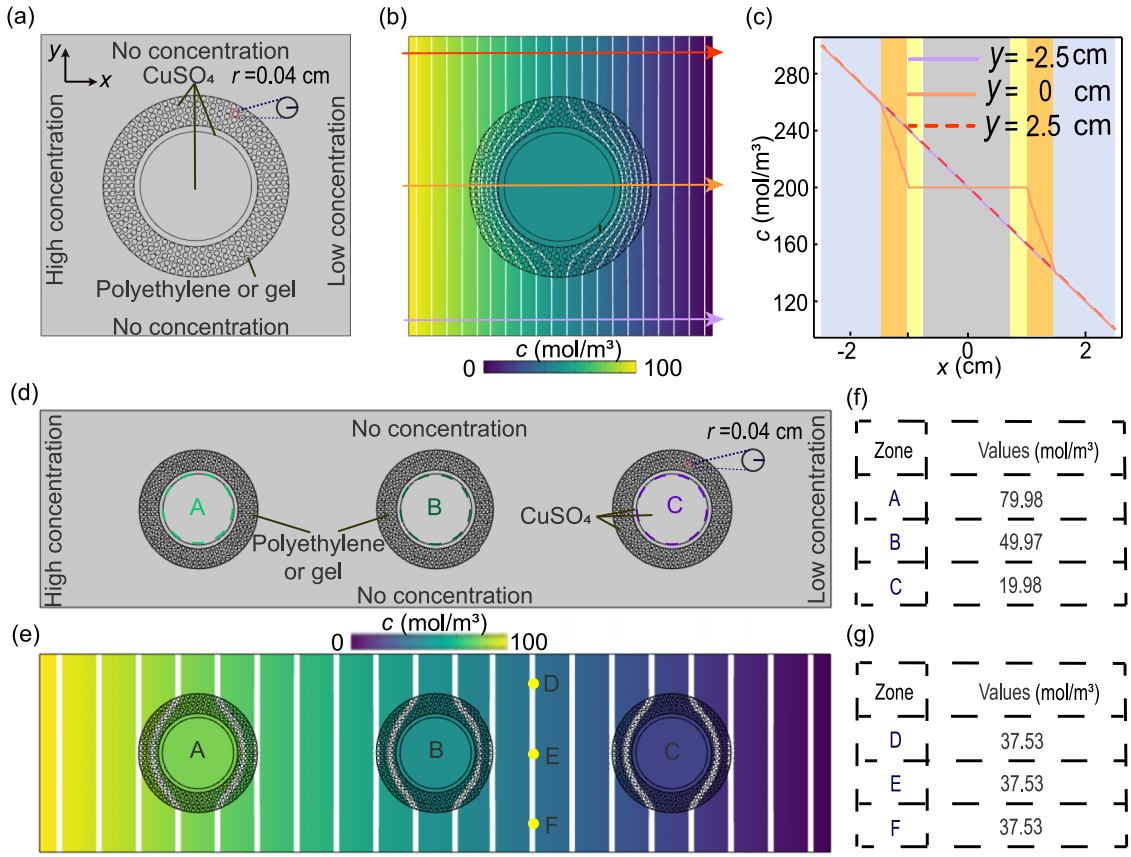

FIG. 4. Experimental suggestions and applications. (a) Schematic of the proposed experimental model. High concentration is applied at the left boundary and low concentration at the right boundary, establishing a concentration gradient along the x axis. The top and bottom boundaries are set as no-flux conditions. The desired concentration distribution in the cloak's outer layer is realized via perforation. (b) Corresponding simulation results for the zero-index cloak. (c) Concentration values extracted from (b) along three horizontal lines at $y = -2.5$, 0, and 2.5 cm. (d) Proposed applicational model illustrating multiple concentration zones. (e) Simulation results showing the background with straight white isoconcentration lines, while zones A, B, and C maintain uniform but distinct concentration values. (f) Concentration values measured within regions A, B, and C, demonstrating clearly separated levels. (g) Concentration values at points D, E, and F, which lie on the same concentration contour in (e), confirming the good cloaking.

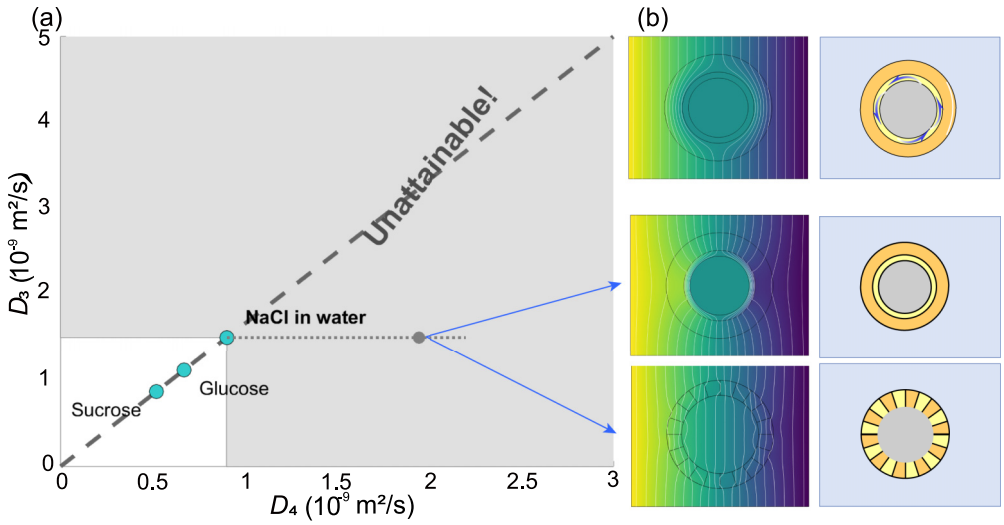

FIG. 5. Material requirements for traditional vs our zero-index-analog cloak. (a) Required outer-layer diffusivity D_3 versus background diffusivity D_4. The requirement for traditional cloaks (dashed line) rises steeply, entering the physically unattainable region for passive materials even for common solutes. (b) Simulation results confirm that our cloak (top) maintains perfect function in a high-diffusivity (NaCl in water) background, while traditional metadevice approaches (middle and bottom) fail, showing significant field distortion.

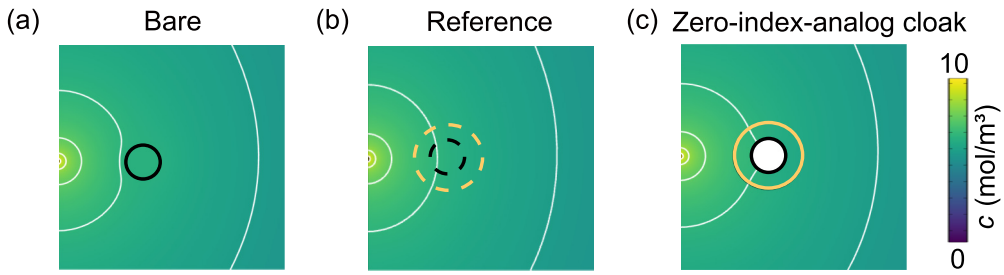

FIG. 6. Steady-state concentration distributions for a point-source-induced nonuniform background field. (a) Bare object without any cloaking structure, showing distortion of the concentration contours due to the presence of the object. (b) Reference case without the object, illustrating the undisturbed background concentration field generated by the point source. (c) Object surrounded by a zero-index-analog cloak, demonstrating that the isotropic cloaking design approximately restores the background concentration field.

the domain. Now consider the boundary conditions. Let region A be a medium approaching the zero-index-analog limit, with diffusivity D_A, and region B a conventional medium. The interface must satisfy concentration continuity $c_A = c_B$ and continuity of normal diffusion flux $\mathbf{n} \cdot D_A(\nabla c)_A = \mathbf{n} \cdot D_B(\nabla c)_B$. To prevent divergence in flux under the limit $D_A \to \infty$, we must have $(\nabla c)_A \to 0$, resulting in a uniform concentration field. This boundary-matching condition directly parallels the optical case with $\mu, \varepsilon \to 0$ and the thermal case with $\kappa \to \infty$.

B. Realizing effective high diffusivity via convection

The central challenge lies in realizing an effectively infinite diffusivity D_2. Since natural solid materials are inherently constrained by the background diffusivity of water, we turn to advection-diffusion dynamics. Specifically, we employ rotational convection within the inner layer to synthesize this extreme parameter. The mass transport is governed by the advection-diffusion equation

$$\frac{\partial c}{\partial t} = D_2 \nabla^2 c - \vec{v}(r) \cdot \nabla c, \tag{4}$$

where $\vec{v}(r)$ is the fluid velocity. To ensure mass conservation, the velocity field should be incompressible and closed. Thus, a natural choice is the circulating field. We consider $v_r = 0$ and $v_\theta = v(r)$ in polar coordinates (r, θ). Under steady-state conditions, Eq. (4) becomes

$$\frac{\partial^2 c}{\partial r^2} + \frac{1}{r}\frac{\partial c}{\partial r} + \frac{1}{r^2}\frac{\partial^2 c}{\partial \theta^2} - \frac{v(r)}{D_2 r}\frac{\partial c}{\partial \theta} = 0. \tag{5}$$

As $v(r)$ approaches infinity, $\partial c/\partial \theta$ approaches zero, resulting in the concentration distribution rotational symmetry. To analyze the radial distribution, we compute the total diffusive flux passing through a circular boundary with radius R, where R lies between R_1 and R_2,

$$Q_R = \int_0^{2\pi} D_2 R \left.\frac{\partial c}{\partial r}\right|_{r=R} d\theta = 2\pi R D_2 \left.\frac{\partial c}{\partial r}\right|_{r=R}, \tag{6}$$

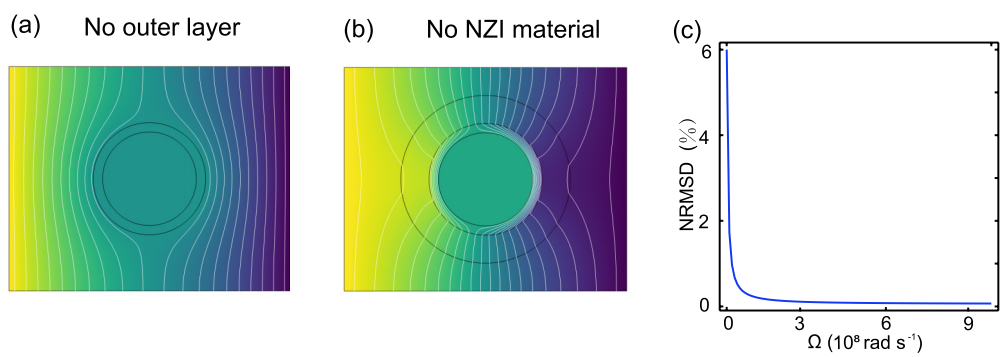

FIG. 7. Component verification and quantitative performance analysis of the concentration cloak. (a) Absence of the outer compensating shell. The object is surrounded only by an inner annular region with rapidly circulating fluid, which provides strong internal homogenization but, without impedance compensation, induces distortions in the background concentration field. (b) Absence of the zero-index-analog effect. Even when the outer shell is present, if the inner fluid remains stationary, the homogenization mechanism is inactive, and the bilayer structure fails to restore the background concentration field, behaving similarly to a diffusive obstacle. (c) Quantitative cloaking performance as a function of angular velocity. The normalized root-mean-square deviation of the background concentration field is plotted versus the angular velocity Ω of the inner circulating flow. The NRMSD is defined as $\langle (c_{\text{cloak}} - c_{\text{ref}})^2 \rangle / \Delta c$, where c_{cloak} and c_{ref} are the concentration fields with and without the cloak, respectively, and Δc is the concentration difference imposed across the system boundaries. The plot shows that the NRMSD rapidly decreases with increasing Ω, signifying improved cloaking performance, and converges to a near-zero value for $\Omega > 3 \times 10^8$ rad/s, indicating the onset of the high-performance regime.

where D_2 is the diffusion coefficient of the species in the fluid. Since there is no internal source, the total flux should be zero. This implies $\partial c/\partial r = 0$, and hence the concentration becomes uniform within the inner layer. Therefore, in a homogenized sense, the rotating fluid mimics the transport response of a solid layer with effectively infinite diffusivity.

Quantitatively, for a solid-body rotation profile $\vec{v} = \Omega r$, asymptotic analysis yields an effective diffusivity to Eq. (2): According to the solution, the effective diffusivity can be expressed as [19]

$$D_2^{\text{eff}} = \frac{R_2^2 + R_1^2}{R_2^2 - R_1^2} D_2 R_2 \sqrt{2\Omega/D_2}. \tag{7}$$

This demonstrates that rapid circulation significantly enhances the effective diffusivity.

C. Unified pseudoconformal mapping and duality

We now employ pseudoconformal mapping to design bilayer cloaks. Crucially, this framework allows us to derive both the proposed zero-index-analog cloak and the traditional impermeable cloak from a unified geometric transformation, revealing a geometric duality between them. The transformation rule of concentration pseudoconformal mapping is given by

$$D_3 = \frac{J D_4 J^T}{\det J}, \tag{8}$$

where J is the Jacobian matrix, $\det J$ is its determinant, and J^T is its transpose. This formulation provides a direct mapping of material parameters between the background and transformed regions, enabling control over the concentration field. The mapping process consists of two steps [see Fig. 2(a)]. First, a circle of radius R_3 in the virtual space is mapped onto an ellipse in the auxiliary space $z_1 = x_1 + iy_1$ with the transformations

$$x_1 = \frac{b}{R_3} x_0, \quad y_1 = \frac{a}{R_3} y_0, \tag{9}$$

where the major and minor axes are given by

$$a = \frac{R_3^2 + R_2^2}{R_3}, \quad b = \frac{R_3^2 - R_2^2}{R_3}. \tag{10}$$

This transformation is pseudoconformal because it preserves the local orthogonality between concentration contours and mass-flux lines, which correspond to the vertical and horizontal grid lines, respectively. The Jacobian matrix corresponding to this geometric transformation is

$$J = \begin{bmatrix} \frac{b}{R_3} & 0 \\ 0 & \frac{a}{R_3} \end{bmatrix}, \tag{11}$$

and the determinant of the Jacobian matrix is

$$\det J = \frac{ab}{R_3^2}. \tag{12}$$

Subsequently, the ellipse in the auxiliary space is mapped onto the annular region $R_2 \leqslant |z| \leqslant R_3$ in physical space $z = x + iy$ via the inverse Zhukovsky conformal transformation

$$z = \frac{z_1 \pm \sqrt{z_1^2 - 4}}{2}. \tag{13}$$

Since this step is a conformal mapping, it does not affect the eigenvalues of the diffusion matrix in the overall mapping. A critical distinction arises from the boundary condition imposed at the inner interface $r = R_2$ in physical space. This boundary corresponds to either an isoconcentration line or a flux line in the virtual space z_0, leading to two distinct types of cloaks.

Case 1: The zero-index-analog cloak (proposed design). We first consider a uniform concentration gradient applied along the x axis [see Fig. 2(b)], for which vertical lines in the virtual space represent concentration contours and horizontal lines indicate flux lines. In this scenario, the effective shell diffusivity is governed by the Jacobian component $J_{xx} = b/R_3$ and the required diffusion coefficient of the outer layer is obtained by calculating the Jacobian transpose and determinant as

$$D_3' = \frac{J_{xx} D_4 J_{xx}{}^T}{\det J_{xx}} = D_4 \frac{R_3^2 - R_2^2}{R_3^2 + R_2^2}. \tag{14}$$

Here the inner boundary $r = R_2$ of the annulus in the physical space is mapped from an isoconcentration line in virtual space; a uniform concentration condition must be ensured at this interface. As discussed in Sec. II B, this behavior is effectively realized in the homogenization limit by employing the active high-diffusivity inner layer ($D_2 \to \infty$) generated by rotational flow [see Fig. 2(f)].

Case 2: The impermeable cloak (dual counterpart). Conversely, if the concentration bias is instead along the y axis [see Fig. 2(c)], our mapping is still angle preserving for the vertical flux lines and horizontal concentration contours in Fig. 2(a). In this orthogonal scenario, the diffusivity is governed by $J_{yy} = a/R_3$. We obtain the diffusion coefficient in the outer layer of the cloak as

$$D_3'' = \frac{J_{yy} D_4 J_{yy}{}^T}{\det J_{yy}} = D_4 \frac{R_3^2 + R_2^2}{R_3^2 - R_2^2}. \tag{15}$$

Here the inner boundary $r = R_2$ of the cloak is mapped from a flux line in the virtual space, implying that a zero normal diffusive flux condition must be imposed at this interface. Physically, this renders the cloaked object completely isolated from diffusive flow, thus achieving perfect impermeability ($D_2 \to 0$) [see Fig. 2(e)].

Duality and practical implications. While methods like scattering cancellation treat impermeable and zero-index cloaks as distinct solutions to separate control goals, our framework reveals a geometric duality between them. This duality stems from the inherent mathematical property of pseudoconformal mapping: the preservation of orthogonality between physical fields. Consequently, the isoconcentration lines in the impermeable cloak correspond exactly to the mass flux streamlines in the zero-index-analog cloak and vice versa [compare the visual evidence in Figs. 2(b) and 2(c)]. Mathematically, swapping the inner-layer boundary condition type (zero flux vs uniform concentration) necessitates an inverse transformation of the outer-layer parameters ($D_3 \propto 1/D_3$), unifying these functionally distinct devices within a

single theoretical framework and offering clearer physical insight for device design. More specifically, this inverse relationship originates from the directional selection of the diagonal components of the Jacobian matrix in the pseudoconformal transformation. For a background concentration gradient applied in the x direction, the effective shell diffusivity is governed by the J_{xx} component, yielding $D_3 \propto b/a$. Conversely, for a gradient applied in the orthogonal y direction, the relevant contribution arises from the J_{yy} component, yielding $D_3 \propto a/b$. These two cases therefore correspond to reciprocal projections of the same diagonal Jacobian matrix, forming a natural mathematical duality rather than two unrelated designs.

This duality also illuminates the distinct nature of the practical limitations inherent in each design. The traditional impermeable cloak operates by forcing flux to detour around the object, which requires an outer-layer diffusivity $D_3 > D_4$. In high-diffusivity backgrounds (e.g., small molecules in water), this constitutes a fundamental material limitation, as finding natural materials with even higher diffusivity is often impossible, a physical dead end. In contrast, our zero-index-analog cloak builds a diffusion bridge, requiring $D_3 < D_4$, a condition robustly satisfied by common porous media or composites. The practical constraint in this case shifts to the inner layer, which must operate in an effectively infinite diffusivity limit ($D_2 \to \infty$). Crucially, unlike the material dead end faced by the traditional cloak, this requirement represents an engineering challenge rather than a material impossibility. As demonstrated above, it can be actively realized via rotational flow in any background medium. Finally, this active bridged interior inherently supports a sensing-while-cloaked functionality, transforming the device from a passive shield into an intelligent interface, a feature fundamentally absent in the isolated interior of traditional cloaks.

III. SIMULATION

The effectiveness of the proposed concentration zero-index-analog cloak was verified through finite-element simulations in COMSOL Multiphysics using the Transport of Diluted Species module. The simulation domain was set as a 30×24 cm^2 rectangle containing a circular cloak with radii $R_1 = 6$ cm, $R_2 = 7$ cm, and $R_3 = 9$ cm. A concentration of 393 mol/m^3 was applied at the left boundary and 293 mol/m^3 at the right boundary, with an initial concentration of 10 mol/m^3 throughout the domain. Figures 7(a) and 7(b) illustrate the critical need for both the active inner layer and the passive shell. Figure 7(a) shows the scenario where the object is enclosed only by the inner zero-index-analog layer (realized by a rapidly circulating fluid). While this layer successfully homogenizes its internal concentration, its near-infinite effective diffusivity creates an impedance mismatch with the background medium. Acting as a strong scatterer, it induces a "short-circuit" effect that drastically distorts the external concentration contours, confirming that the NZI-analog layer alone cannot achieve cloaking in this configuration. Conversely, Fig. 7(b) shows the bilayer structure without the zero-index-analog effect (i.e., the internal fluid is stationary, $\Omega = 0$). In this case, the condition of $D_2 \to \infty$ is violated, and the structure fails to function as a cloak, behaving similarly to a diffusive obstacle. Thus, both the homogenizing active layer and the compensating outer shell are essential for the device's performance. These control groups confirm that neither component suffices in isolation; the homogenization provided by the active core and the impedance matching provided by the passive shell must work in concert.

Steady-state concentration distributions are shown in Figs. 2(d)–2(f). Figure 2(d) presents the reference case without any object, serving as the baseline for comparison. The diffusion coefficient for the homogeneous background was set to 2×10^{-9} m^2/s. Figure 2(e) shows the simulated result of a traditional bilayer cloak [10] in a high-diffusivity background. Theoretically, cloaking under such conditions would require the outer layer to have a diffusivity of 8.13×10^{-9} m^2/s. Realizing such high diffusivity is practically challenging using common naturally occurring materials in this context, leading to noticeable distortions in the background concentration field. In contrast, Fig. 2(f) demonstrates that the zero-index-analog cloak maintains an undisturbed background concentration profile even under the same high-diffusivity conditions. In this case, the inner layer's effective diffusivity is 7.31×10^{-4} m^2/s, significantly higher than that of the background 2×10^{-9} m^2/s, corresponding to an equivalent flow speed of $\Omega = 1000$ rad/s. According to pseudoconformal mapping theory, the outer-layer diffusivity is calculated as 0.49×10^{-9} m^2/s, which is easily attainable. It is important to address the practical implications of implementing such high angular velocities. An estimation of the Reynolds number ($\mathrm{Re} = \rho v L/\mu$) helps clarify the flow regime. For the dimensions in our simulation $R_1 = 6$ cm and $R_2 = 7$ cm and assuming water as the fluid ($\rho \approx 1000$ kg/m^3 and $\mu \approx 10^{-3}$ Pa s), the characteristic velocity at the outer boundary is $v = \Omega R_2 \approx 70$ m/s for $\Omega = 1000$ rad/s, and the characteristic length is $L = R_2 - R_1 = 1$ cm. This yields a Reynolds number of $\mathrm{Re} \approx 7 \times 10^5$, a value that clearly indicates the flow would be turbulent. Nevertheless, this transition to turbulence is expected to enhance, rather than undermine, the device's performance. The core mechanism of the inner layer is the rapid homogenization of concentration, and the efficient eddy mixing inherent in turbulence is a far more potent homogenization mechanism than simple laminar advection. Therefore, the model used in our analysis [Eq. (4)], which assumes laminar flow, serves as a conservative and theoretically tractable approximation for elucidating the fundamental physics. The presence of turbulence in a physical realization would likely strengthen the cloaking effect. This conclusion is consistent with studies on analogous thermal cloaks, where turbulent flow models have confirmed the robustness of the cloaking phenomena [19]. Finally, Fig. 2(g) quantitatively compares cloaking performance along the x axis, where different line colors represent different scenarios. The close match between the zero-index-analog cloak curve and the reference case confirms the proposed zero-index cloaking strategy's effectiveness in achieving high-performance concentration cloaking in high-diffusivity environments.

Transient simulations also were conducted using COMSOL Multiphysics, employing the same model and parameters as the steady-state simulations. Figures 3(a)–3(d) display the concentration profiles at 500 s with a bare object, a refer-

ence case, a traditional bilayer cloak, and a zero-index-analog cloak, highlighting the differences between traditional bilayer cloaks and our design. In a traditional bilayer cloak, the concentration in the cloaked region changes slowly due to the isolating nature of its inner boundary condition (zero flux) and is unaffected by variations in the background concentration, making it difficult to detect changes in the surrounding environment. As a result, such designs reveal their inherent limitations in dynamically changing environments. In contrast, the zero-index-analog cloak introduces a high-speed rotating fluid into the inner layer, endowing it with an extremely high effective diffusivity. This not only ensures a uniform concentration field in the cloaked region but also enables it to respond rapidly to external changes, offering significant advantages for applications requiring rapid environmental adaptability. To quantify this transient behavior, Fig. 3(e) presents the concentration evolution over time at the model's center. The zero-index-analog cloak exhibits a rapid concentration change, closely matching the bare object case, unlike the traditional bilayer cloak that responds more slowly, highlighting the device's potential as a concentration sensor capable of dynamically responding to environmental variations. Moreover, quantitative analysis at $x = 10$ cm shows that the zero-index-analog cloak better preserves the background concentration distribution, exhibiting a lower root-mean-square deviation of the concentration contours compared to the traditional cloak. Ideally, a perfect cloak would eliminate background concentration deviations, indicating full restoration of the external field. Compared to traditional bilayer cloaks, the zero-index-analog cloak better approaches this ideal, exhibiting better cloaking performance in concentration fields [see Fig. 1(b)]. To quantitatively evaluate the cloak's performance as a function of the angular velocity Ω, we calculated the normalized root-mean-square deviation (RMSD) of the background concentration field. The detailed results, plotted in Fig. 7(c), confirm that the distortion of the concentration field decreases dramatically with increasing angular velocity, reaching a near-perfect cloaking effect at high Ω.

The zero-index-analog cloak is not limited to circular objects but can in principle apply to objects of arbitrary shapes. We numerically verified its cloaking performance for triangular and elliptical objects. Steady-state simulation results demonstrate that the cloak effectively restores the background concentration field, as shown in Figs. 3(f) and 3(h). Similar performance is expected for other object geometries. Figures 3(g) and 3(i) present the corresponding velocity field distributions, where $|v|$ denotes speed and red arrows indicate flow direction.

IV. EXPERIMENTAL SUGGESTION

A specific experimental design was proposed to further assess the feasibility of the theoretical model. The proposed experimental setup is shown in Fig. 4(a). The system consists of a square domain with a side length of 5 cm. A circular object with a radius of 1 cm is placed at the center, surrounded by a bilayer structure with inner and outer radii of 1.1 and 1.5 cm, respectively. The concentrations at the left and right boundaries are fixed at 100 and 0 mol/m^3, respectively. The top and bottom boundaries are set as no-flux boundaries. The diffusion coefficient in both the background and the object region is set to 1.27×10^{-9} m^2/s. A blue $CuSO_4$ solution is used as a tracer to visualize the diffusion field. According to theoretical calculations, the required diffusion coefficient for the outer layer is 0.38×10^{-9} m^2/s. However, it is difficult to find a single natural material whose diffusion coefficient precisely matches this value. Therefore, we adopt the effective-medium theory. Specifically, we uniformly disperse microstructures made of polyethylene or hydrogel [29] (materials with much lower diffusivity) into the $CuSO_4$ background solution to tune the overall effective diffusivity of the outer layer to meet the design requirement. For the inner layer, we propose a high-flow strategy based on a rotating disk. Driven by a motor, the disk generates strong circular flow, effectively approximating a high-diffusivity medium and significantly accelerating concentration homogenization. For a practical implementation, the circulating fluid would be physically confined within a sealed annular channel (from $r = R_1$ to $r = R_2$) constructed from a low-permeability material such as poly(methyl methacrylate) or glass. This robust physical barrier serves two critical functions: It isolates the active fluidic region from both the central cloaked core and the external environment and it prevents mass leakage. The viability of this physical confinement strategy for active fluidic components is supported by successful demonstrations in analogous thermal metamaterial experiments [19,25]. Furthermore, potential extraneous perturbations, such as flow instabilities or mechanical vibrations, can be mitigated with standard engineering practices. For instance, precision motor control can ensure consistent flow velocity, and vibration-damping mounts can be used to isolate the motor, ensuring that such nonideal effects do not compromise the cloaking performance.

We further performed simulations to validate the feasibility of the proposed experimental design, as presented in Fig. 4(b). The isoconcentration lines in the background remain straight, while the concentration inside the object region is uniform and approximates the average value of the left and right boundaries. These results confirm that the cloak enables the object to sense external concentration changes without disturbing the background field. To quantitatively evaluate the cloaking performance, we extracted concentration values along the x axis at three different positions ($y = -2.5$, 0, and 2.5 cm), as shown in Fig. 4(c). The background concentration distributions at three positions are identical, confirming that the cloak effectively restores the background diffusion field. The concentration inside the object remains uniform, with the gradient approaching zero.

V. DISCUSSION AND CONCLUSION

In this work we established a theoretical framework that functionally emulates the behavior of zero-index media [30–34] in mass diffusion. By exploiting the mathematical isomorphism between the Helmholtz equation and the diffusion equation, we identified that the active homogenization of the inner layer plays the role of the near-zero-index medium in wave physics. To mitigate the disturbances that the active layer induces in the surrounding concentration field, we incor-

porated pseudoconformal mapping theory, enabling the design of an isotropic concentration cloak operating in a zero-index-analog regime. Numerical simulations demonstrated that this cloak exhibits excellent invisibility and sensitive response for various target shapes, even in high-diffusion environments. Furthermore, an experimental scheme combining perforated structures and effective-medium theory offers a solid platform for validating the theoretical design. This work extends near-zero-index-inspired concepts to mass diffusion [35], providing a unified and practically accessible framework for designing controllable diffusion metadevices with potential applications in biomedicine, environmental protection, and industrial diffusion management [36–38].

It is important to clarify the distinction between the theoretical principle and its physical implementation. While the zero-index concept requires an effectively unbounded diffusivity, strictly passive scattering-cancellation cloaks face an insurmountable bottleneck: Naturally occurring solids generally cannot exceed the diffusivity of the aqueous background. We explicitly acknowledge that the mechanism of enhancing effective diffusivity via fluid rotation is physically well established in fluid dynamics. However, the contribution of this work lies in the strategic integration of this mechanism into transformation optics to overcome the aforementioned material barrier. Consequently, our design represents a necessary paradigm shift from identifying nonexistent passive materials to synthesizing active metamaterials. By strictly mapping the infinite-diffusivity requirement to the rotational Péclet number, we bridge the gap between hydrodynamic transport and metamaterial design. This approach is not merely a compromise, but the viable physical realization pathway for diffusive zero-index cloaking in aqueous environments, offering additional benefits such as active tunability (e.g., via flow speed adjustment) that remain inaccessible to static passive systems.

Furthermore, a key consideration is the extension of our two-dimensional (2D) framework to three dimensions. The effectiveness of our pseudoconformal approach, particularly its ability to yield isotropic parameters, is deeply rooted in the unique flexibility of 2D complex analysis. In stark contrast, 3D transformations are severely constrained by Liouville's theorem, which limits conformal maps to a narrow class of Möbius transformations, precluding the design of cloaks for arbitrarily shaped objects. Despite this fundamental limitation, our method remains highly relevant for quasi-3D applications possessing translational symmetry. Our 2D design can be straightforwardly extruded along the invariant axis to create 3D devices, such as a cloak for a long cylinder or a concentrator for a planar channel, for controlling fluxes primarily in the 2D plane. The design of truly 3D, isotropic diffusion metamaterials for arbitrary objects, however, remains a profound open challenge. This likely requires new theoretical paradigms beyond direct coordinate transformations and thus represents a fertile ground for future investigation.

The practical feasibility of the zero-index cloak for potential applications is further demonstrated by finite-element simulations. The simulated system is a rectangle measuring 20 cm in length and 5 cm in width. We placed three circular objects in the domain, with centers at $(-6, 0)$, $(0, 0)$, and $(6, 0)$, and each object has a radius of 1 cm. Each object is surrounded by a bilayer cloak with inner and outer radii of 1.1 and 1.5 cm, respectively. The boundary conditions are consistent with those used in the simulation for the experimental suggestion [see Fig. 4(d)]. Figure 4(e) presents the simulated concentration distribution, where the three object regions each form a stable microenvironment with distinct concentration levels, ordered as $A > B > C$ [see Fig. 4(f)], indicating that each bilayer cloak successfully enables independent concentration control. Meanwhile, the simulation results show that the concentration contours in the background remain perfectly straight and undisturbed. The extracted concentration values at three background points D, E, and F in Fig. 4(e) are exactly the same, as shown in Fig. 4(g), further confirming that the background field remains fully preserved. These results demonstrate that the designed cloak enables independent concentration control in multiple regions without mutual interference. Furthermore, the active nature of our design, criticized as a complexity in passive contexts, here emerges as a critical functional advantage. Unlike traditional impermeable cloaks that isolate the interior (creating a blind spot), our fluid-mediated bridge maintains information exchange. This strictly enforces the field continuity required for sensing-while-cloaked functionality, transforming the device from a mere shield into a noninvasive probe, a capability fundamentally inaccessible to static insulating cloaks. In fields like systems biology or chemical sensing, this ability to decouple an object's physical disturbance from its environmental awareness enables experimental studies that are impossible with conventional designs. This capability provides a foundation for applications such as multicell coculture and high-throughput drug screening. First, on multicellular coculture platforms [39–43], the cloak can maintain multiple stable concentration microenvironments. It enables independent or coculture of cell combinations such as tumor and normal cells, neurons and glial cells, and stem cells alongside their differentiated derivatives. This capability provides a robust tool for analyzing how nutrient or drug concentration affects specific cellular behaviors, for resolving biological gradient effects, and for facilitating disease mechanism studies, cell function evaluation, and precision medicine. Second, in high-throughput screening [44], the device can efficiently generate multiple stable concentration gradients in parallel. This enables rapid dose-response analysis and toxicity assessment, significantly improving the throughput of toxicological screening and drug sensitivity testing [45] and supporting research in personalized medicine and quantitative toxicology. Finally, the device can mimic chronic low-concentration drug exposure in tissues or infection lesions. It creates a stable environment for long-term drug sensitivity and resistance evolution studies, helping to elucidate mechanisms of pathogen resistance and guiding strategies for antibiotic therapy [46].

The proposed active zero-index-analog strategy significantly enhances mass diffusion efficiency, overcoming conventional material constraints. This advancement opens new possibilities for precise manipulation of mass transport [47,48] in various high-diffusivity applications. Moreover, the zero-index cloak lays the foundation for developing intelligent diffusion metamaterials [49–52]. Looking ahead, a key

challenge is to incorporate environmental responsiveness for adaptive control in dynamic environments [53]. For example, integrating materials sensitive to temperature [54], *p*H, or specific biomolecules would allow the cloak's properties to be dynamically tuned in response to changing biological conditions. This would significantly enhance the robustness and applicability of such devices in practical scenarios.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

APPENDIX A: COMPARATIVE ANALYSIS OF MATERIAL PARAMETER REQUIREMENTS FOR TRADITIONAL AND ZERO-INDEX-ANALOG DESIGNS

It is well established that effective anisotropy can be approximated by layering isotropic materials. Our motivation, however, is not to challenge this principle but to reveal a fundamental material-parameter bottleneck inherent to all traditional passive cloaks, whether they are implemented as anisotropic layered composites or as simplified isotropic shells. This bottleneck becomes insurmountable in high-diffusivity backgrounds.

Let us first analyze the layered anisotropic cloak. According to transformation theory, a cloak must guide flux around a central region. To achieve this, the effective tangential diffusivity $D_{\theta,\mathrm{eff}}$ must be greater than the background diffusivity D_4, as it needs to facilitate faster transport along the circumference. Using effective-medium theory, $D_{\theta,\mathrm{eff}} = f_A D_A + f_B D_B$. For this arithmetic mean to exceed D_4, at least one of the constituent materials (say, material A) must have a diffusivity D_A that is also greater than D_4. When the background itself is a high-diffusivity medium (e.g., NaCl in water), finding a passive material with an even higher diffusivity to serve as a constituent layer becomes physically impossible. This constraint fundamentally limits the applicability of layered cloaks to low-diffusivity environments.

Next consider the simplified isotropic bilayer cloak, which represents another major class of traditional designs. The required diffusivity of its outer layer (D_3) is given by

$$D_3 = \frac{R_3^2 + R_2^2}{R_3^2 - R_2^2} D_4, \tag{A1}$$

where R_2 and R_3 are the inner and outer radii of the layer. Crucially, the geometric factor is always greater than 1, mandating that $D_3 > D_4$. This requirement is identical in nature to the one faced by the layered anisotropic cloak. Therefore, both cornerstone designs of traditional passive cloaking share the same fatal flaw: They fundamentally require a material component with a diffusivity exceeding that of the background.

Figure 5 plots these two relationships and highlights a critical bottleneck: As the background medium becomes more diffusive, the material requirement for the traditional cloak rapidly enters a physically unattainable region for any known passive materials. In stark contrast, our zero-index-analog approach relies on an effective transport enhancement mechanism rather than an intrinsic material property. As a result, the required shell diffusivity increases much more slowly with the background diffusivity and remains well within the range achievable by passive composite materials, even in highly diffusive environments. This analysis thus clarifies that our primary motivation is to overcome the dual challenges of traditional diffusive metamaterials: the fundamental paradox of their anisotropic implementation and the extreme material parameter demands of their simplified isotropic counterparts.

APPENDIX B: ROBUSTNESS OF THE ISOTROPIC APPROXIMATION AGAINST FIELD NONUNIFORMITIES

It is essential to clarify the theoretical scope of our isotropic design. The derivation of a scalar diffusivity D_3 is predicated on a pseudoconformal transformation that is mathematically exact for a uniform background gradient. From a first-principles perspective, a perfect transformation cloak for an arbitrary and nonuniform field (e.g., one generated by a proximal point source) would require an anisotropic and inhomogeneous shell. Therefore, our isotropic design should be understood as an effective approximation in nonuniform fields. This approximation proves to be robust. Simulations of bare, reference, and cloaked cases are shown in Figs. 6(a), 6(b), and 6(c), respectively. The cloaking performance remains excellent provided the background field is locally uniform on the scale of the device, i.e., when the characteristic length scale of the field's spatial variation is significantly larger than the cloak's radius. This robust tolerance to moderate field nonuniformities enhances the practical utility of our simplified isotropic design for real-world applications. Compared to the cloaked and reference cases, the bare object under a nonuniform background field introduces distortions in the concentration distribution.

APPENDIX C: QUANTITATIVE PERFORMANCE ANALYSIS OF THE CONCENTRATION CLOAK

To quantitatively evaluate the influence of the angular velocity Ω on the cloak's performance [transitioning from the state in Fig. 7(b) to an ideal cloaking condition], we conducted a parametric study. The cloaking performance is quantified by the normalized root-mean-square deviation (NRMSD) of the concentration field in the external background region, defined as

$$\mathrm{NRMSD} = \frac{\langle (c_{\mathrm{cloak}} - c_{\mathrm{ref}})^2 \rangle^{1/2}}{\Delta c}, \tag{C1}$$

where $\langle (c_{\mathrm{cloak}} - c_{\mathrm{ref}})^2 \rangle$ denotes the spatial average over the background domain, c_{cloak} and c_{ref} are the concentration fields

with and without the cloak, respectively, and Δc is the total concentration difference across the system boundaries.

Figure 7(c) plots the NRMSD as a function of the angular velocity Ω. The results clearly demonstrate that for a stationary fluid ($\Omega = 0$), the NRMSD is significant, indicating substantial distortion of the background field. As Ω increases, the strong advection dramatically enhances the effective diffusivity of the inner layer, leading to a rapid improvement in cloaking performance, manifested by a sharp decrease in the NRMSD value. For Ω exceeding approximately 3×10^8 rad s^{-1} in the numerical model, the NRMSD saturates at a very low value, indicating that a near-perfect cloaking effect has been achieved. This analysis not only validates the effectiveness of our design but also establishes a direct relationship between its performance and the key operational parameter, providing crucial guidance for practical implementation and energy consumption optimization.

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